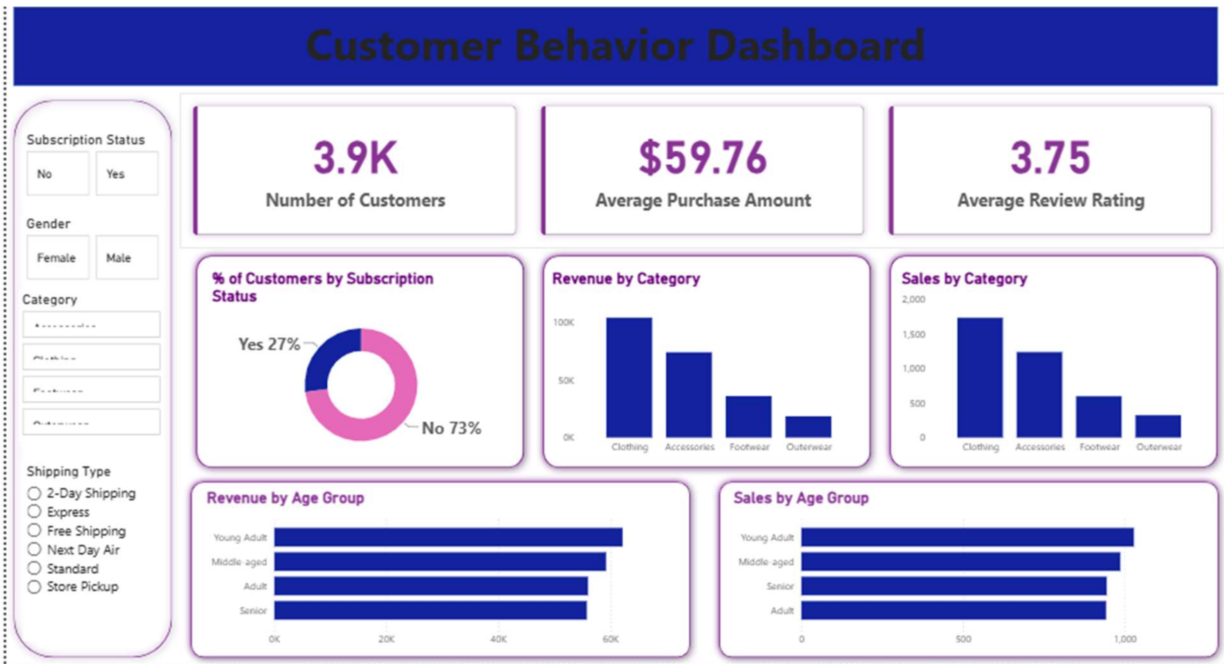
Customer Behavior Analytics Case Study

Transforming Customer Shopping Data into Business Insights Using SQL, Python & Power BI



Business Context

Retail companies collect millions of customer transactions that contain valuable information about purchasing behavior, product preferences, spending patterns, and customer loyalty. However, raw customer data alone does not provide meaningful business value. To support strategic decisions, organizations need to understand the hidden patterns behind customer actions and transform transactional data into actionable insights.

In this project, I simulated the role of a Data Analyst working with a retail company that wanted to better understand its customers' shopping behavior in order to improve sales performance, customer satisfaction, and long-term loyalty.

The company had observed changes in customer purchasing patterns across demographics, product categories, and shopping preferences. However, management lacked visibility into the key factors influencing customer decisions, repeat purchases, and revenue contribution.

The main business challenge was:

How can consumer shopping data be leveraged to identify customer trends, improve engagement, and optimize marketing and product strategies?

To answer this question, I developed an end-to-end customer analytics workflow that transformed raw customer shopping data into structured analysis, business insights, and decision-support recommendations.

Phase 1: Understanding Customer Behavior Data

The project started with a raw customer shopping dataset containing transaction-level information. Before performing any analysis, the first step was understanding how customer behavior is represented within business data.The objective was to understand how customer characteristics, purchasing decisions, and business outcomes were connected.

The initial investigation focused on several questions:

How are customers and transactions connected?
Which customer attributes influence purchasing decisions?
What factors contribute to higher revenue generation?
How can customer data be transformed into meaningful business knowledge?

The dataset included important customer behavior variables such as demographics, age groups, gender, purchased products, product categories, purchase amounts, discount usage, customer reviews, shipping preferences, subscription status, and previous purchase history.

At this stage, the focus was not only understanding the data structure but also identifying which factors could explain customer engagement, spending behavior, and loyalty.

Phase 2: Data Preparation and SQL Database Analysis

After understanding the dataset, the next step was transforming raw customer records into a structured analytical environment.

In real-world organizations, analysts rarely work directly with raw files. Business data usually comes from operational systems and must be organized before meaningful analysis can be performed.Therefore, the dataset was prepared and structured using SQL to create a reliable analytical foundation.

The process involved organizing customer transaction data, creating structured tables, preparing analytical datasets, and developing SQL queries to answer important business questions.

SQL was selected as the main analytical tool because it represents one of the core skills used by Data Analysts to extract insights from business databases.

Through this stage, the project moved from raw customer records into a structured environment where customer behavior could be measured, compared, and analyzed.

Phase 3: Understanding Customer Revenue Contribution

The first analytical objective was understanding which customer groups contribute the most revenue.

The business question was:

How much revenue is generated by male and female customers?

To answer this question, customer spending behavior was analyzed by gender.

The analysis measured revenue contribution and compared purchasing patterns between different customer groups using SQL aggregation techniques such as:

- SUM()
- GROUP BY

This analysis helped identify customer segments that generate higher revenue and provided information that can support targeted marketing strategies.

Phase 4: Understanding the Impact of Discounts on Customer Spending

Discount campaigns are commonly used by retailers to increase sales, but discounts do not always create valuable customers.

The next business question was:

Which customers used discounts but still generated above-average purchase values?

The analysis focused on identifying customers who received discounts while maintaining higher-than-average spending behavior.

SQL subqueries and aggregation functions were used to compare individual customer purchases against the average purchase amount.

This analysis helped evaluate whether discount strategies were attracting valuable customers or simply reducing profitability.

Phase 5: Product Performance and Customer Satisfaction Analysis

Customer reviews provide important signals about product quality and customer satisfaction.

The project analyzed product performance by asking:

Which products have the highest average customer review ratings?

The analysis calculated average product ratings and identified products with stronger customer satisfaction levels.

SQL techniques included:

- AVG()
- GROUP BY
- ORDER BY
- LIMIT

These insights can help businesses identify successful products, improve product positioning, and make better inventory decisions.

Phase 6: Understanding Shipping Behavior and Customer Spending

Customer experience is influenced not only by products but also by delivery options.

The analysis investigated:

Do customers using Standard or Express shipping demonstrate different purchasing behavior?

The project compared average purchase amounts across shipping preferences.

Using SQL aggregation techniques, customer spending patterns were analyzed based on delivery choices.

These insights help businesses understand whether premium shipping options are associated with higher-value customers and whether delivery strategies influence customer purchasing behavior.

Phase 7: Subscription Analysis and Customer Loyalty

Customer loyalty programs are important tools for improving retention and increasing customer lifetime value.

The business question was:

Do subscribed customers provide higher business value than non-subscribers?

The analysis compared subscribers and non-subscribers based on:

- Average spending
- Total revenue contribution
- Number of customers

SQL functions such as:

- COUNT()
- AVG()
- SUM()
- GROUP BY

were applied to measure customer value.

The results provide insights into whether subscription programs successfully encourage stronger customer relationships.

Phase 8: Understanding Discount Dependency by Product

Different products respond differently to promotional campaigns.

The analysis explored:

Which products have the highest percentage of purchases using discounts?

The project calculated discount usage rates for each product by applying SQL conditional logic.

Techniques included:

- CASE statements
- SUM()
- COUNT()

This analysis helps identify products that depend heavily on promotions and supports smarter pricing and marketing strategies.

Phase 9: Customer Segmentation Based on Purchasing History

Customer segmentation allows businesses to personalize communication and improve retention.

Customers were classified based on previous purchase behavior into:

- New customers
- Returning customers
- Loyal customers

This segmentation was created using SQL CASE statements, CTEs, and aggregation techniques.

The goal was to understand customer maturity and identify groups requiring different engagement strategies.

These insights support personalized marketing campaigns and customer retention planning.

Phase 10: Product Ranking and Category Performance

Retail organizations need to understand which products drive category success.

The business question was:

What are the top-performing products within each category?

To answer this, products were ranked based on purchase frequency using SQL window functions.

Techniques applied included:

- ROW_NUMBER()
- PARTITION BY
- Window Functions

This analysis helps businesses identify popular products, optimize inventory planning, and improve product placement strategies.

Phase 11: Repeat Customer and Subscription Relationship

Customer loyalty is often connected with subscription behavior.

The project investigated:

Are repeat customers more likely to become subscribers?

Customers with higher previous purchase frequency were analyzed and compared based on subscription status.

This analysis provided insights into whether loyal customers are more likely to engage with membership programs.

Phase 12: Revenue Contribution Across Customer Demographics

Understanding demographic spending patterns helps businesses improve targeting strategies.

The final analysis examined:

Which age groups contribute the highest revenue?

Revenue contribution was calculated across different customer age groups using SQL aggregation.

These insights support demographic-based marketing decisions and help organizations better understand their most valuable customer segments.

Advanced SQL Skills Demonstrated

Throughout this project, SQL was used as a business analytics tool rather than only a querying language.

The project demonstrated practical experience with:

- Data aggregation using SUM(), COUNT(), and AVG()
- Customer segmentation using CASE statements
- Advanced filtering using subqueries
- Customer grouping using CTEs
- Ranking analysis using ROW_NUMBER()
- Business KPI development
- Revenue and customer behavior analysis

Final Business Questions Answered

Through this analysis, customer transaction data was transformed into answers for important business questions:

How much value does each customer segment generate?

Which customers represent the highest revenue opportunities?

Do discounts attract valuable customers?

Which products create stronger customer satisfaction?

Do subscription customers generate higher lifetime value?

Which products drive category performance?

Which demographic groups contribute the most revenue?

How can customer behavior insights improve marketing and retention strategies?

Project Outcome

This project demonstrates a complete Customer Analytics workflow, starting from raw customer transaction data and progressing through data preparation, SQL analysis, customer behavior modeling, and business intelligence reporting.

The analytical journey was:

Raw Customer Data
↓
Data Preparation and Structuring
↓

SQL Database Analysis
↓
Customer Behavior Insights
↓
Power BI Visualization
↓
Business Strategy Recommendations

The project demonstrates how raw customer transactions can be transformed into meaningful business intelligence that supports revenue growth, customer retention, and data-driven decision-making.